

TIØ4145 Corporate Finance

Exercise 4: Estimating the cost of capital, investor behavior and capital market efficiency

Deadline: Monday September 22nd, 2025, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday September 18th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on September 15th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapters 12 and 13.

Tasks

Chapter 12 – Estimating the Cost of Capital

1. You need to estimate the equity cost of capital for XYZ Corp. You have the following data available regarding past returns:



Year	Risk-free Return	Market Return	XYZ Return
2017	3%	7%	8%
2018	1%	-43%	-37%

- What was XYZ's average historical return?
- Compute the market's and XYZ's excess returns for each year. Estimate XYZ's beta using the excess returns. (If you want to calculate the beta in Excel, use the SLOPE/STIGNINGSTALL function).
- Estimate XYZ's historical alpha. (If you want to calculate the alpha in Excel, use the INTERCEPT/SKJÆRINGSPUNKT function).

- d. Suppose the current risk-free rate is 4%, and you expect the market's return to be 8%. Use the CAPM to estimate an expected return for XYZ Corp's stock.
 - e. Would you base your estimate of XYZ's equity cost of capital on your answer in part (a) or in part (d)? How does your answer to part (c) affect your estimate? Explain.

2. In mid-2012, Rite Aid had CCC-rated, 6-year bonds outstanding with a yield to maturity of 17.3%. At the time, similar maturity Treasuries had a yield of 3%. Suppose the market risk premium is 6% and you believe Rite Aid's bonds have a beta of 0.31. The expected loss rate of these bonds in the event of default is 58%.
 - a. What annual probability of default would be consistent with the yield to maturity of these bonds in mid-2009?
 - b. In mid-2015, Rite-Aid's bonds had a yield of 6.6%, while similar maturity Treasuries had a yield of 1.6%. What probability of default would you estimate now?

3. Unida Systems has 40 million shares outstanding trading for \$10 per share. In addition, Unida has \$100 million in outstanding debt. Suppose Unida's equity cost of capital is 15%, its debt cost of capital is 8%, and the corporate tax rate is 40%.
 - a. What is Unida's unlevered cost of capital?
 - b. What is Unida's after-tax debt cost of capital?
 - c. What is Unida's weighted average cost of capital?

4. You would like to estimate the weighted average cost of capital for a new airline business. Based on its industry asset beta, you have already estimated an unlevered cost of capital for the firm of 9%. However, the new business will be 24% debt financed, and you anticipate its debt cost of capital will be 6%. If its corporate tax rate is 32%, what is your estimate of its WACC?

Exam Question – 2019

Easteros Inc., an unlevered firm, has expected earnings before interest and taxes of \$2 million per year. Easteros' tax rate is 40%, and the market value is \$12 million. The stock has a β of 1, and the risk-free rate is 9%. The risk premium is 6%. Management is considering the use of perpetual debt (the size of the firm would remain constant). The default free interest rate on debt is 12%.

What is the equity cost of capital and WACC before the debt is issued?

Chapter 13 – Investor Behavior and Capital Market Efficiency

1. Allison and Bill are both mutual fund managers, although Allison is more skilled than Bill. Both have \$300 million in assets under management and charge a fee of 1%/year. Allison is able to generate a 2% alpha before fees and Bill is able to generate a 1% alpha before fees.
 - a. What is the alpha investors earn in each fund (that is, the alpha after fees are taken out)?
 - b. Which fund will experience an inflow of funds?
 - c. Assume that both managers have exhausted the supply of good investment opportunities and so they will choose to invest any new funds received in the market portfolio and so those funds will earn a zero alpha. How much new capital will flow into each fund?
 - d. Once the new funds have stopped flowing, what is the alpha before and after fees of each fund? Which fund will be larger?
 - e. Calculate each manager's compensation once the capital has stopped flowing. Which manager has the highest compensation?
2. What are the only conditions under which the market portfolio might not be an efficient portfolio?
3. Consider the following stocks, all of which will pay a liquidating dividend in a year and nothing in the interim. Assume the risk-free rate is 3% and the market risk premium is 7%.

	Market Capitalization (\$ million)	Expected Liquidating Dividend (\$ million)	Beta
Stock A	800	1000	0.77
Stock B	750	1000	1.46
Stock C	950	1000	1.25
Stock D	900	1000	1.07

- a. Calculate the expected return of each stock and the correlation between the expected returns and market capitalization of the stocks.
- b. What does the CAPM predict the expected return for each stock should be?
- c. Clearly, the CAPM predictions are not equal to the actual expected returns so the CAPM does not hold. You decide to investigate this further. To see what kind of mistakes the CAPM is making, you decide to regress the actual expected return onto the expected return predicted by the CAPM. What is the intercept and slope coefficient of this regression?

- d. What are the residuals of the regression in (c)? That is, for each stock, compute the difference between the actual expected return and the best fitting line given by the intercept and slope coefficient in (c).
- e. What is the correlation between the residuals you calculated in (d) and market capitalization?

Exam Question – 2018

Assume that the CAPM is valid, and that the stocks pay no dividends in the next period. You are given the following information:

	$r_{\text{free rate}}$	$E[R_{\text{Market}}]$	$E[R_{\text{Stock}}]$	β	Price_0	$E[\text{Price}_1]$
A	10%	20%	10%		100	
B	10%	20%		1.5		125
C	10%	20%	20%			240

Complete the table by calculating the missing data.

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.